

MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust ®, Mangalore)

Autonomous Institute Affiliated to V.T.U, Belagavi, Approved by AICTE, New Delhi

Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution



PROJECT PHASE – 1 REPORT

“Edge-Optimized ML Framework for Realtime Camouflaged Target Detection in Tactical Military Surveillance Systems”

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

by

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

2025-26

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CERTIFICATE

This is to certify that the project work entitled “**EDGE-OPTIMIZED ML FRAMEWORK FOR REALTIME CAMOUFLAGED TARGET DETECTION IN TACTICAL MILITARY SURVEILLANCE SYSTEMS**” is a Bonafide work carried out by **CHAITHANYA R RAO (4MT23AI014), GOURAV (4MT23AI024), VINUSH KUMAR (4MT23AI061), YAKSHITH K D (4MT23AI062)** in partial fulfillment for the award of degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning of Mangalore Institute of Technology & Engineering (Autonomous)** during the year **2025-26**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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Chapter 4

4. PROPOSED METHODOLOGY

4.1 Proposed Approach and Algorithm:

The proposed system adopts a Deep Gradient Network (DGNet) [18] as its core detection architecture. DGNet is specifically designed for Camouflaged Object Detection and addresses the fundamental limitation of standard detectors — their reliance on color and texture features that camouflage explicitly mimics. DGNet decouples the detection process into two complementary streams: Contextual Awareness and Texture Gradient Identification, enabling the system to effectively detect camouflaged targets even when thermal signatures are suppressed or unavailable.

The detection framework operates as follows:

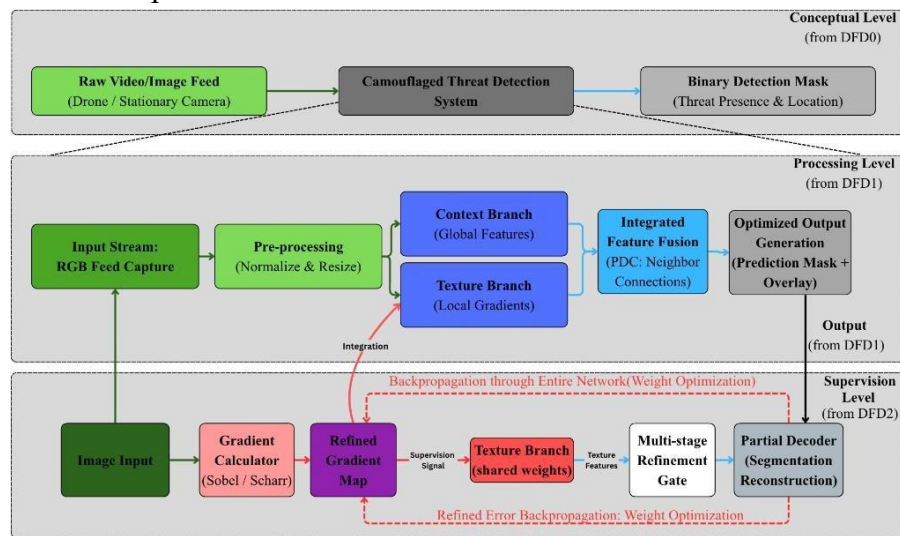


Fig 4.1 DGNet Camouflaged Threat Detection

- **Context Branch (The Searcher):** Uses a deep backbone (EfficientNet or ResNet) to capture global semantic features, answering the question: Where is the general area of interest?
- **Texture Branch (The Identifier):** A specialized branch that monitors Gradient-Induced Transitions (GIT) [18]. It acts as a structural probe to find the unnatural edges of military hardware, rifles, or personnel by supervising the network to identify sharp changes in pixel intensity.
- **Partial Decoder Component (PDC):** A unified decoder aggregating multi-level features from both branches using neighbor connections to preserve high-resolution spatial details, resulting in a precise binary detection mask [18].

4.3 System design and Workflow:

4.3.1 Data flow design:

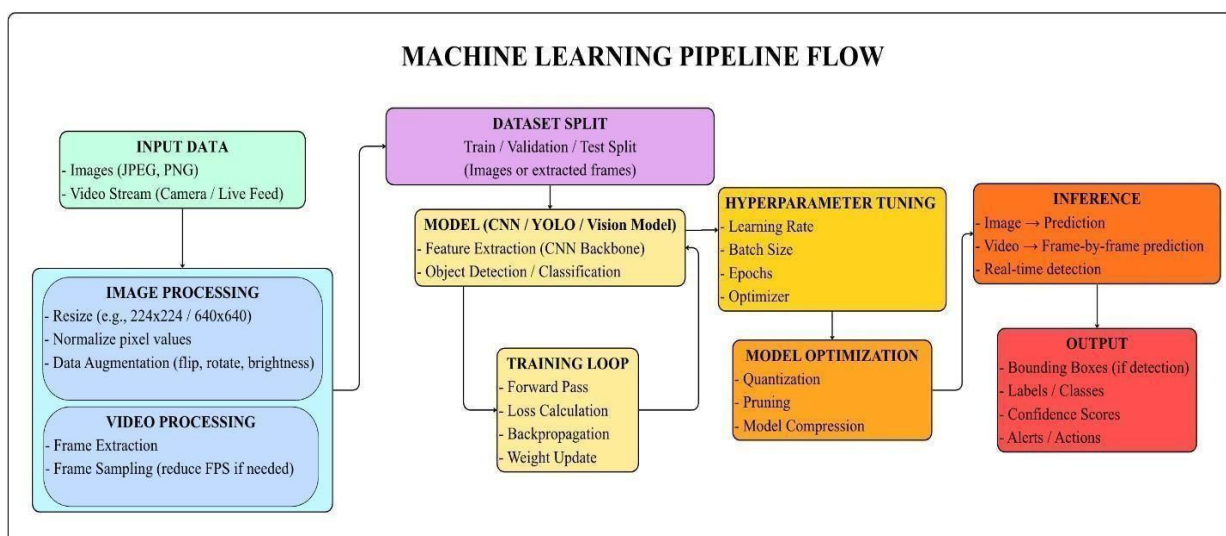


Fig 4.2 Data flow design

The data flow operates as a sequential machine learning pipeline, progressing from raw data ingestion to optimized real-time inference.

- **Data Ingestion** (The "Input Stage"): Processes static images and live video streams through dimensional resizing, normalization, spatial augmentation, and frame extraction.
- **Model Training** (The "Learning Core"): Partitions data into subsets and trains the feature extraction architecture, optimizing accuracy via hyperparameter tuning and iterative weight updates.
- **Real-time Inference** (The "Deployment Engine"): Compresses the network using quantization and pruning for edge devices, executing frame-by-frame detection to generate bounding boxes and automated alerts.

4.3.2 Deployment Design: Cloud vs. Edge:

For tactical military use, the project prioritizes **Edge Deployment**:

- **Edge Hardware**: NVIDIA Jetson Orin or specialized AI accelerators on edge devices or raspberry pi 5
- **Optimization**: The model is converted to **ONNX** or **TensorRT** formats to achieve **80+ FPS**.
- **Architecture**: A containerized approach using **Docker** ensures that the AI service can be updated over the air (OTA) in the field.
- **API Integration**: A lightweight FastAPI server is used to communicate detection coordinates to the Command and Control (C2) interface.